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UNIT 1 — MINI PROJECT LAB REPORT

**Link Budget and Coverage • FDD vs TDD • 0G–6G
Comparison**

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Repository: <https://github.com/IRAFASHA-Justine/MOBILE-COMMUNICATION-WORKS->

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Introduction

This report covers the three Mobile Communications Unit 1 mini projects: (1.1) a link-budget and coverage-range simulation, (1.2) a comparison of FDD and TDD resource utilization under asymmetric traffic, and (1.3) validation and extension of the 0G–6G generation comparison. Each project follows the structure requested by the lecturer: title, objective, problem statement, tools and software, background, procedure, observations, conclusion and references.

The procedures and requirements are taken from the Unit 1 lab manual. All numerical results in Projects 1.1 and 1.2 come from the Python programs written for this report (their printed output and CSV files), and all figures are the charts those programs saved. Project 1.3 results come from the recommender program's saved test file and from the external sources listed in the References.

Software environment and repository

All programs were written in Python 3 and run from the terminal in Visual Studio Code. The complete source code, CSV result files and charts are in the GitHub repository given on the title page. The repository is organised as follows:

MOBILE_COMMUNICATIONS_MINI_PROJECTS/

```
01_Link_Budget/ link_budget.py, results.csv, task3_fading_results.csv,  
                link_budget_comparison.png, three_band_comparison.png,  
                capacity_vs_distance.png, README.md  
02_FDD_TDD/    fdd_tdd_simulator.py, results.csv, guard_period_results.csv,  
                fdd_tdd_comparison.png, guard_period_comparison.png, README.md  
03_0G_6G/      generation_recommender.py, generation_research.csv,  
                research_table.xlsx, test_results.txt
```

Mini Project 1.1 — Simulating Link Budgets and Coverage Range with Python

Project Title

Mini Project 1.1: Simulating Link Budgets and Coverage Range with Python (700 MHz low-band versus 28 GHz millimetre-wave).

Objective

To translate the link-budget equation and the free-space path-loss formula of Section 1.10 into working Python code, and to use it to compare received power and maximum usable range for a low-band (700 MHz) and a millimetre-wave (28 GHz) carrier at equal transmit power. The project also extends the basic model with antenna gain, a mid-band carrier, a fading margin, Shannon capacity, and a power-versus-bandwidth comparison.

Problem Statement

A radio signal weakens as it travels, and the loss also grows with carrier frequency. A receiver can decode a signal only if the received power stays above its sensitivity with some margin. The experiment asks: under one fixed set of transmitter, antenna and loss assumptions, at what distance does the received power fall below a -100 dBm sensitivity threshold for 700 MHz and for 28 GHz, and why do the two bands differ so strongly? Six extension tasks then test how antenna gain, a third band, a fading margin, SNR/capacity, and the choice between more power or more bandwidth change the result.

Tools and Software Used

- Python 3 (script `link_budget.py`), run from the VS Code terminal.
- NumPy for vectorised calculation; Matplotlib for plotting; the csv module (standard library) for saving results.
- Git and GitHub for version control and submission. No SDR or radio hardware was required: the experiment is a pure software simulation.

Background

The link budget adds all gains and subtracts all losses between transmitter and receiver (all values in dB or dBm):

$$P_{RX} = P_{TX} + G_{TX} - L_{path} + G_{RX} - L_{other}$$

Path loss is estimated with the simplified free-space formula $L_{path} \text{ (dB)} = 20 \cdot \log_{10}(d_{km}) + 20 \cdot \log_{10}(f_{MHz}) + 32.44$. Because it depends on the logarithm of both distance and frequency, loss rises by 20 dB for every tenfold increase in distance and by $20 \cdot \log_{10}(f_2/f_1)$ dB when the carrier frequency is raised from f_1 to f_2 . The link margin is P_{RX} minus the receiver sensitivity. For capacity, the Shannon–Hartley theorem gives $C = B \cdot \log_2(1 + \text{SNR})$, which grows linearly with bandwidth but only logarithmically with SNR.

Table 1.1: Fixed simulation parameters

Parameter	Value
Transmit power P_{TX}	40 dBm
Transmit antenna gain G_{TX}	15 dBi (24 dBi in Task 1 for 28 GHz)
Receive antenna gain G_{RX}	0 dBi
Other losses L_{other}	3 dB (11 dB in Task 3)

Parameter	Value
Receiver sensitivity	−100 dBm
Reference distance	1 km
Distance range	500 log-spaced points from 0.01 km to 100 km (np.logspace(-2, 2, 500))
Bands	700 MHz, 3.5 GHz (Task 2), 28,000 MHz
Noise floor / bandwidth (Task 4)	−104 dBm for a 10 MHz channel

Procedure Used

1. Import NumPy and Matplotlib and write the function `fspl_db(d_km, f_mhz)` for free-space path loss.
2. Write `link_budget_dbm(p_tx, g_tx, l_path, g_rx, l_other)` that returns received power in dBm.
3. Create 500 logarithmically spaced distances from 0.01 km to 100 km.
4. Fix the parameters of Table 1.1 so that only frequency differs between the 700 MHz and 28 GHz curves.
5. Compute path loss and P_{RX} at every distance for both bands.
6. Plot both curves on a log-distance axis with the dashed −100 dBm sensitivity line, axis labels and a legend; save the chart.
7. Use boolean indexing to find the largest distance where P_{RX} exceeds sensitivity, and compute path loss, P_{RX} and link margin at the 1 km reference distance; save results.csv.
8. Repeat with the Lab Tasks: 24 dBi gain at 28 GHz (Task 1); add 3.5 GHz and plot three bands (Task 2); add an 8 dB fading margin and save `task3_fading_results.csv` (Task 3); compute SNR and Shannon capacity for $B = 10$ MHz (Task 4); compare doubling power with doubling bandwidth (Task 5); write the TDD duty-cycle note (Task 6).

Exact Observations

A. Main comparison: 700 MHz versus 28 GHz

Table 1.2: Summary at the 1 km reference distance (from results.csv)

Band	Freq. (MHz)	Path loss at 1 km (dB)	P_{RX} at 1 km (dBm)	Margin (dB)	Max range in simulation	Max range, analytical check
700 MHz	700	89.34	−37.34	62.66	≥100 km (edge of plotted range)	≈1,358 km
28 GHz	28,000	121.38	−69.38	30.62	33.66 km	≈34.0 km

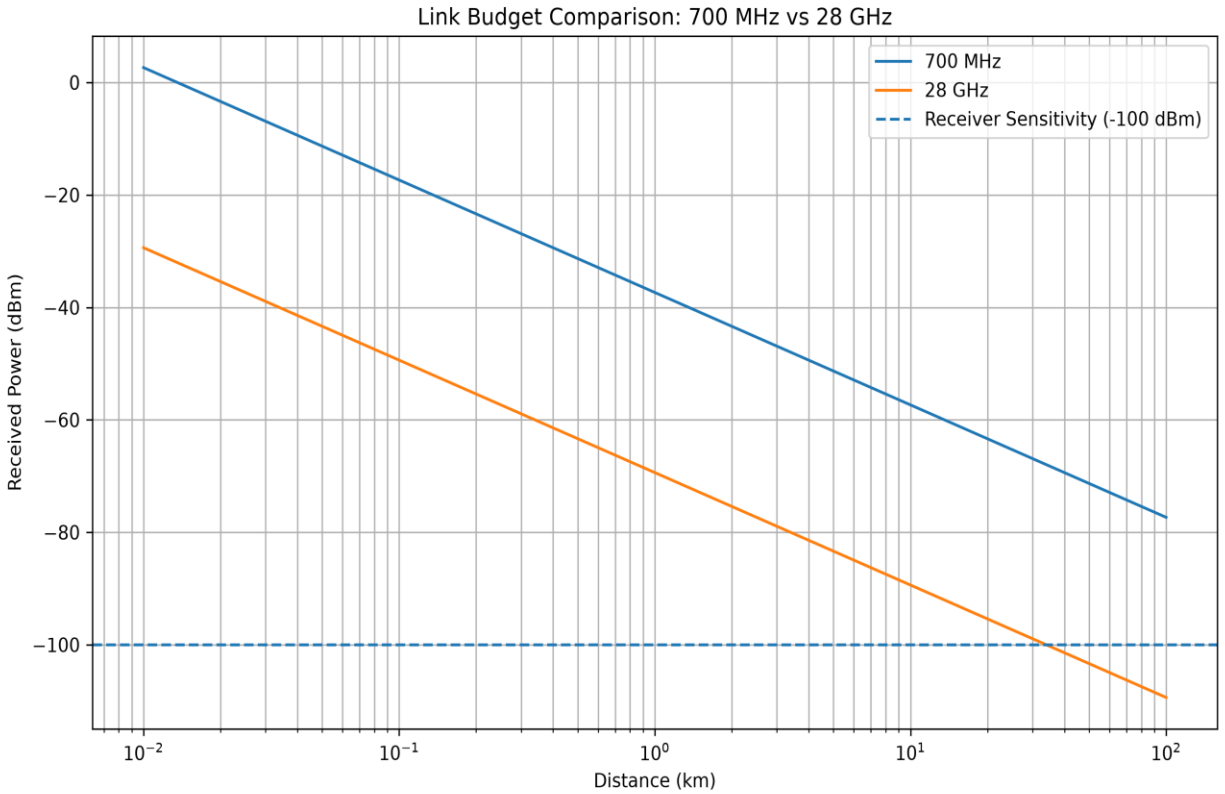


Figure 1.1: Received power versus distance for 700 MHz and 28 GHz with the -100 dBm sensitivity line (link_budget_comparison.png)

Both curves fall as straight lines on the log-distance axis, which confirms that the path-loss formula is linear in $\log_{10}(d)$. At 100 km, 700 MHz is still at about -77 dBm, well above the -100 dBm threshold, whereas 28 GHz is at about -109 dBm and has already crossed it, at 33.66 km. The 700 MHz curve never crosses the threshold inside the plotted 0.01–100 km range, so the simulated value of 100 km for 700 MHz is only the edge of the distance array, not a true limit. Solving the link budget directly for the distance at which $P_{RX} = -100$ dBm (maximum path loss 152 dB) gives about 1,358 km for 700 MHz and 34.0 km for 28 GHz; the 28 GHz value agrees with the simulation.

B. Written analysis: why the range differs

Only frequency differs between the two curves, so the entire difference comes from the term $20 \cdot \log_{10}(f_{\text{MHz}})$ in the free-space formula. Going from 700 MHz to 28,000 MHz adds $20 \cdot \log_{10}(28000/700) = 20 \cdot \log_{10}(40) \approx 32$ dB of path loss at every distance (89.34 dB versus 121.38 dB at 1 km). Since path loss grows by 20 dB per decade of distance, a 32 dB penalty is equivalent to reducing the range by a factor $10^{(32/20)} = 40$. The 700 MHz link therefore reaches about 40 times farther than the 28 GHz link, which is between one and two orders of magnitude as expected in the lab manual and matches the “low-band for coverage, mmWave for capacity” trade-off of Sections 1.5 and 1.10.

The manual also suggests that millimetre-wave should reach “at most a few hundred metres”. The simulation gives 33.66 km because free-space loss is the best-case lower bound: it ignores walls, foliage, rain, ground reflection and a path-loss exponent larger than 2. The numbers should therefore be read as upper bounds for comparing bands, not as real deployment ranges.

C. Lab Tasks 1–3: antenna gain, third band and fading margin

Table 1.3: Results of Tasks 1, 2 and 3

Task	Result from the program	Interpretation
1	28 GHz range rises from 33.66 km to 94.61 km ($\times 2.81$) when G_{TX} is raised from 15 to 24 dBi.	The extra 9 dB of gain equals $10^{(9/20)} \approx 2.82$ times more range. It recovers a large part of the loss but 28 GHz still falls short of 700 MHz.
2	3.5 GHz stays above -100 dBm over the whole plotted range (analytical limit ≈ 272 km).	Its curve lies between the others: $20 \cdot \log_{10}(5) \approx 14$ dB below 700 MHz and $20 \cdot \log_{10}(8) \approx 18$ dB above 28 GHz (Figure 1.2).
3	With +8 dB fading margin: 28 GHz range falls from 33.66 km to 13.37 km. 700 MHz and 3.5 GHz stay at ≥ 100 km in the plotted range (analytical ≈ 541 km and ≈ 108 km).	8 dB corresponds to $10^{(-8/20)} \approx 0.40$, so every band loses about 60 % of its range. Only 28 GHz shows the shrinkage because the others exceed the 100 km plot limit.

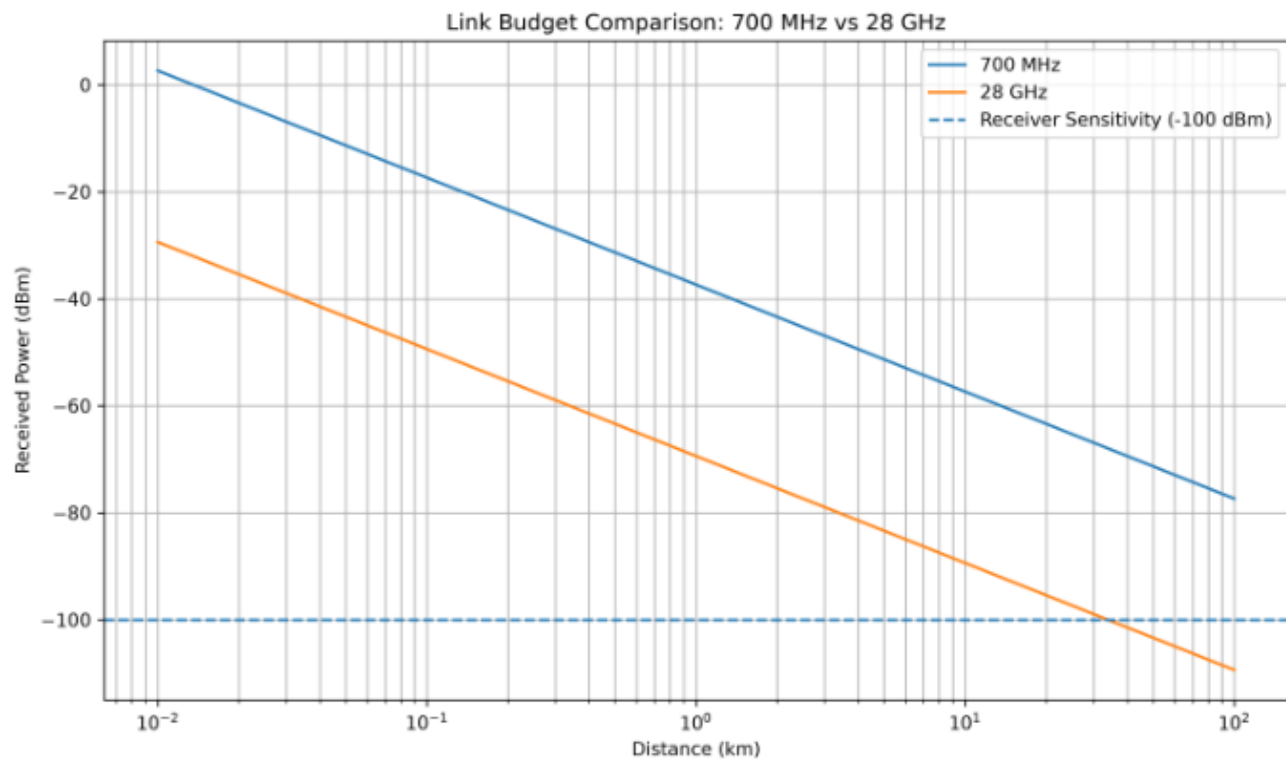


Figure 1.2: Received power for 700 MHz, 3.5 GHz and 28 GHz with the sensitivity line

D. Task 4: SNR and Shannon capacity

With a fixed noise floor of -104 dBm (10 MHz channel), $SNR = P_{RX} + 104$ dB, and the capacity ceiling is $C = B \cdot \log_2(1 + SNR)$ with $B = 10$ MHz.

Table 1.4: SNR and capacity ceiling versus distance (computed with the program's formulas)

Distance	700 MHz SNR (dB)	700 MHz C (Mbps)	3.5 GHz SNR (dB)	3.5 GHz C (Mbps)	28 GHz SNR (dB)	28 GHz C (Mbps)
0.01 km	106.7	354.3	92.7	307.9	74.6	247.9
1 km	66.7	221.4	52.7	175.0	34.6	115.0
10 km	46.7	155.0	32.7	108.6	14.6	49.0
100 km	26.7	88.6	12.7	42.9	-5.4	3.7

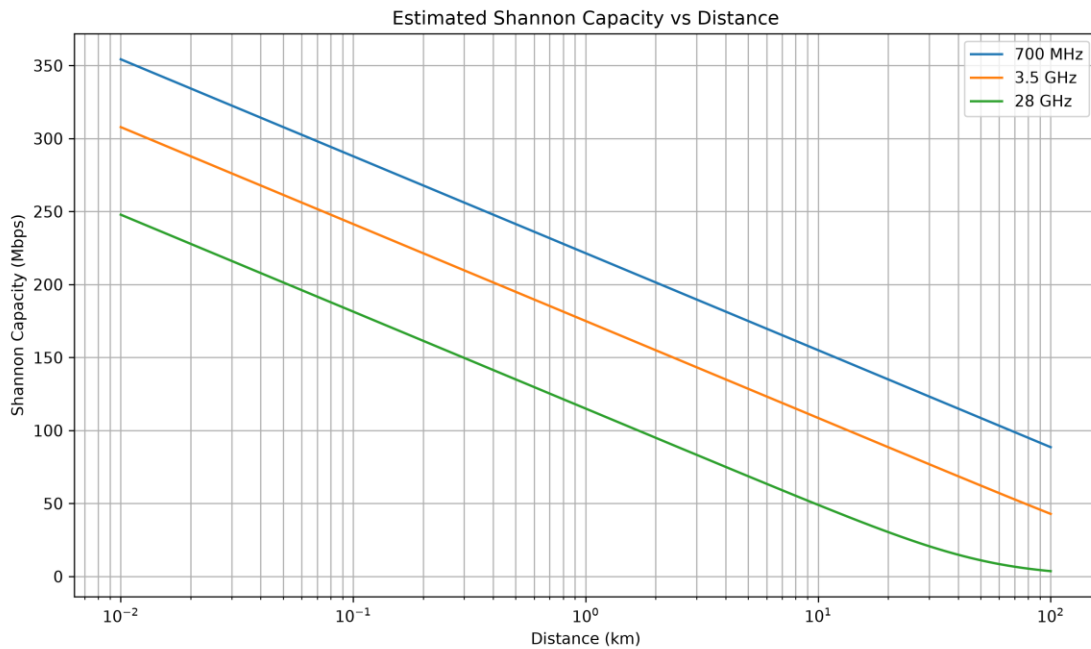


Figure 1.3: Estimated Shannon capacity versus distance for the three bands, $B = 10$ MHz
(capacity_vs_distance.png)

P_{RX} falls by a constant 20 dB for every tenfold increase in distance, but capacity does not fall in the same way. While SNR is high, each 20 dB lost removes only about 6.6 bit/s/Hz (roughly 66 Mbps at 10 MHz), so 700 MHz capacity drops from 354 Mbps to 89 Mbps over four decades of distance. When SNR becomes small, the logarithm turns almost linear in SNR and capacity collapses: 28 GHz falls to 3.7 Mbps at 100 km, where SNR is already negative (−5.4 dB). Capacity therefore degrades slowly at short range and very quickly near the edge of coverage.

E. Task 5: doubling transmit power versus doubling bandwidth

Table 1.5: 3.5 GHz link at 1 km, SNR ≈ 52.7 dB, $B = 10$ MHz

Case	Capacity (Mbps)	Change
Baseline	174.99	—
Doubling transmit power (+3 dB)	184.96	+5.7 %
Doubling bandwidth (noise floor kept at −104 dBm, as in the lab assumption)	349.99	+100 %
Doubling bandwidth with noise power also doubled (−101 dBm)	≈ 330	$\approx +89$ %

Doubling power adds 3 dB to SNR, which adds only about 1 bit/s/Hz inside the logarithm (about 10 Mbps here). Doubling bandwidth multiplies the whole expression, so capacity scales linearly with B . Even when the larger noise bandwidth is taken into account (last row), the gain is still close to double. This is the logarithmic-versus-linear scaling of Section 1.10, and it explains why operators prefer wide mmWave channels to simply transmitting more power.

F. Task 6: TDD duty cycle versus FDD (qualitative)

A link budget normally assumes that the transmitter is active whenever data is sent. An FDD system transmits continuously on its own paired channel in each direction, so the duty cycle per direction is 100 %. A TDD system shares one channel: each direction transmits only in its own slots, and guard periods carry no payload. For example, a 70:30 DL:UL split with 5 % guard time gives roughly 66.5 % downlink

and 28.5 % uplink duty cycle. The power received during a slot is unchanged, but the average throughput and average transmitted energy per second are lower than for continuous FDD, and the short uplink duty cycle limits the average uplink data rate of a phone.

G. Program output

Figure 1.4: Terminal output of link_budget.py (summary, Tasks 1–6)

```
=====
MINI PROJECT 1.1 - LINK BUDGET SIMULATION
=====

Band: 700 MHz
Frequency: 700 MHz
Path loss at 1 km: 89.3420 dB
Received power at 1 km: -37.3420 dBm
Link margin at 1 km: 62.6580 dB
Maximum usable range: 100.0000 km
-----
Band: 28 GHz
Frequency: 28000 MHz
Path loss at 1 km: 121.3832 dB
Received power at 1 km: -69.3832 dBm
Link margin at 1 km: 30.6168 dB
Maximum usable range: 33.6553 km
-----
█
```

Conclusion

The simulation shows that frequency and distance both control received power, and that the free-space loss at 28 GHz is 32 dB larger than at 700 MHz, so the 700 MHz link reaches about 40 times farther (more than 100 km in the plotted range, $\approx 1,358$ km analytically, versus 33.66 km). Extra antenna gain restores part of the mmWave range ($\times 2.81$ with 24 dBi), a fading margin shortens every band by about 60 %, and capacity falls gently with distance until SNR becomes low and then collapses. Increasing bandwidth is far more effective than increasing power. Because free-space propagation is a best case, the results are valid for comparing bands but not as exact real-world coverage predictions.

Mini Project 1.2 — Comparing FDD and TDD Resource Utilization Under Traffic Asymmetry

Project Title

Mini Project 1.2: Comparing FDD and TDD Resource Utilization Under Traffic Asymmetry.

Objective

To build a simplified simulator that models how FDD and TDD allocate uplink and downlink capacity, and to quantify how the effective throughput and utilization of each scheme respond as traffic becomes more downlink-heavy, including the effect of the TDD guard-period overhead.

Problem Statement

Section 1.8 explains that FDD reserves fixed, symmetric capacity on two paired channels, while TDD shares one channel in time and can rebalance the uplink-to-downlink slot ratio, at the cost of guard periods and network synchronization. Real mobile traffic is heavily downlink-dominated. For a fixed total capacity and a given degree of traffic asymmetry, how much usable capacity does each scheme actually deliver, and how does this change as asymmetry grows and as the guard overhead increases?

Tools and Software Used

- Python 3 (script `fdd_tdd_simulator.py`), run from the VS Code terminal.
- NumPy for calculation, pandas for tabulating and saving results (`results.csv`, `guard_period_results.csv`), and Matplotlib for the comparison charts.
- Git and GitHub for submission. No physical mobile-network hardware is required.

Background

FDD keeps separate uplink and downlink frequency channels continuously available, separated by a duplex gap, so it needs paired spectrum but no network-wide timing. TDD uses one unpaired channel and divides time between downlink and uplink; short guard periods are inserted at each direction change. TDD therefore needs tight synchronization between neighbouring cells and loses a small part of the frame to guard time, but its DL/UL ratio can be adjusted to follow the traffic. This is why TDD dominates 5G NR mid-band and mmWave deployments, where spectrum is usually unpaired and traffic is downlink-heavy.

Model used in the simulator (capacity expressed in arbitrary units, total = 100):

- **FDD:** two fixed channels of 50 units each. Delivered traffic in each direction = $\min(\text{demand}, 50)$. Capacity left unused in one direction cannot be moved to the other.
- **TDD:** one channel of 100 units. A guard fraction g of the frame carries no traffic, so usable capacity = $100 \cdot (1 - g)$. The DL/UL time split is set equal to the demand ratio, so each direction has capacity $(1 - g) \cdot 100 \cdot \text{share}$ and delivers $\min(\text{demand}, \text{capacity})$.
- **Demand and utilization:** offered traffic equals the total capacity (100 units) split as DL:UL. Utilization = useful traffic delivered $\div$ total capacity available (100). A guard of 5 % corresponds to one guard slot in twenty.

Table 2.1: Simulation parameters

Parameter	Value
Total capacity	100 units
FDD capacity per direction	50 units (downlink) and 50 units (uplink)
TDD guard overhead (main run)	5 % (one guard slot in twenty)
TDD guard overhead (sensitivity runs)	2 %, 5 % and 10 %
Traffic scenarios (DL:UL)	50:50, 70:30, 80:20 and 90:10

Procedure Used

1. Define the total capacity (100 units), the fixed FDD capacity per direction (50 + 50) and the TDD guard fractions.
2. Implement `simulate_fdd()`: delivered traffic per direction is limited by its own fixed 50-unit capacity.
3. Implement `simulate_tdd()`: usable capacity is reduced by the guard fraction and the DL/UL split follows the demand ratio.
4. Run both models for the 50:50, 70:30, 80:20 and 90:10 scenarios with a 5 % guard; store throughput per direction, total throughput and utilization in `results.csv`.
5. Plot FDD and TDD utilization for every scenario.
6. Repeat the TDD calculation with guard fractions of 2 %, 5 % and 10 % (`guard_period_results.csv`) and plot the result.
7. Compare TDD and FDD in every scenario, derive the break-even asymmetry, and interpret the trade-offs of Section 1.8.

Exact Observations

A. FDD versus TDD (5 % guard)

Table 2.2: Effective throughput per direction (units) and utilization (from `results.csv`)

DL:UL demand	FDD DL	FDD UL	FDD utilization	TDD DL	TDD UL	TDD utilization	TDD – FDD
50:50	50.0	50.0	100 %	47.5	47.5	95 %	–5 points
70:30	50.0	30.0	80 %	66.5	28.5	95 %	+15 points
80:20	50.0	20.0	70 %	76.0	19.0	95 %	+25 points
90:10	50.0	10.0	60 %	85.5	9.5	95 %	+35 points

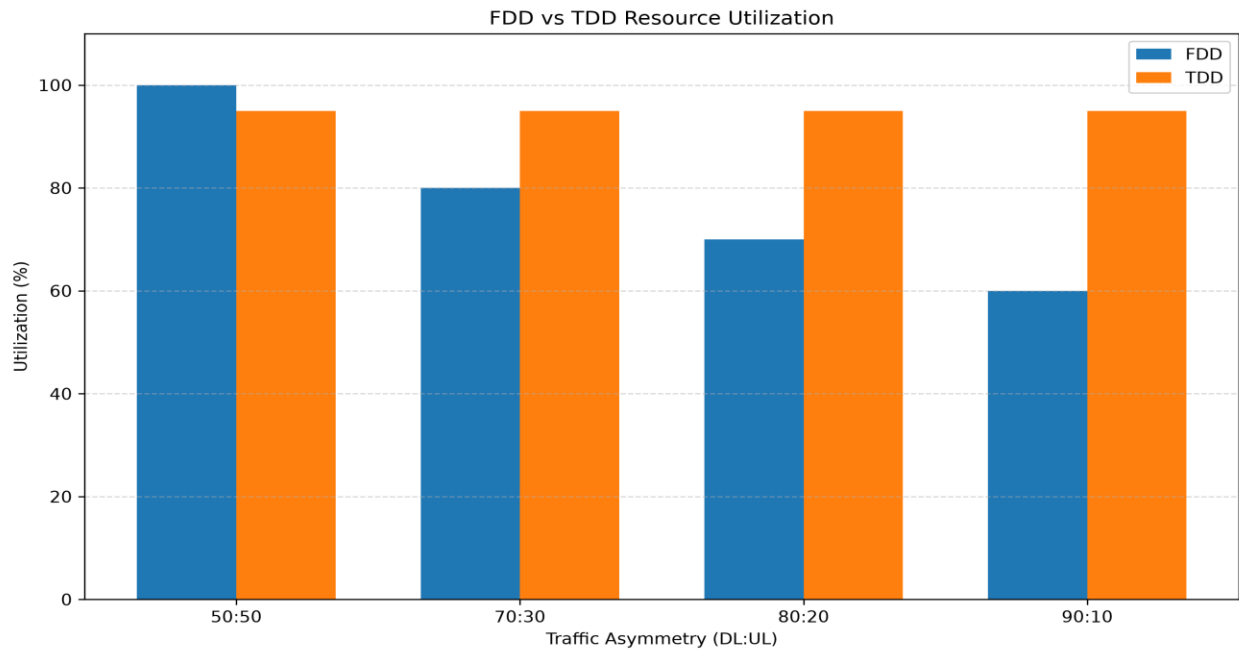


Figure 2.1: FDD versus TDD resource utilization for each traffic scenario, 5 % TDD guard (fdd_tdd_comparison.png)

FDD utilization falls from 100 % to 80 %, 70 % and 60 % as traffic becomes more downlink-heavy. At 90:10 the FDD downlink is saturated at 50 units (40 units of demand cannot be served) while 40 of the 50 uplink units are idle and cannot be lent to the downlink. TDD stays at 95 % in every scenario because its time split follows the demand; the 5 % gap from 100 % is the guard overhead. The value 95 % is specific to this simulation and is not a universal TDD figure.

B. Effect of the guard period

Table 2.3: TDD utilization for different guard fractions (from guard_period_results.csv)

Scenario	FDD	TDD, 2 % guard	TDD, 5 % guard	TDD, 10 % guard
50:50	100 %	98 %	95 %	90 %
70:30	80 %	98 %	95 %	90 %
80:20	70 %	98 %	95 %	90 %
90:10	60 %	98 %	95 %	90 %

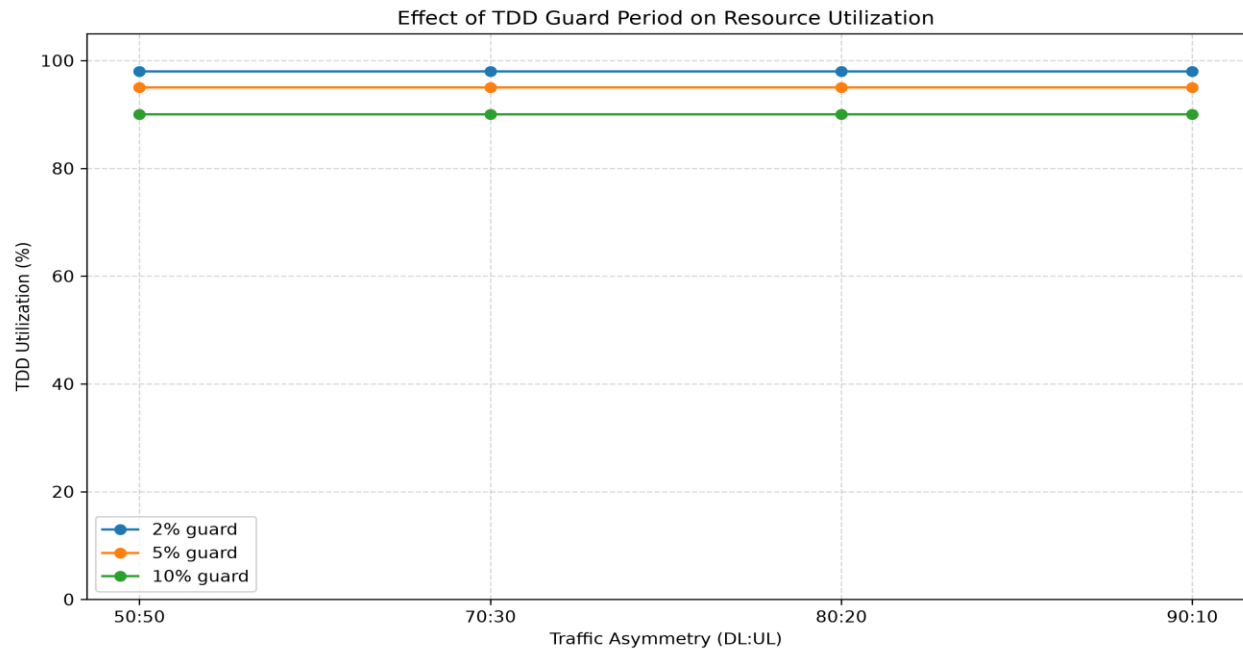


Figure 2.2: TDD utilization for 2 %, 5 % and 10 % guard overhead (guard_period_comparison.png)

In this model the guard overhead lowers TDD utilization by exactly the guard fraction, independently of the traffic ratio, so the three curves are horizontal. A larger overhead lowers the line and therefore reduces TDD's advantage over FDD.

C. Break-even asymmetry and where FDD stays competitive

With downlink share d , FDD utilization is $100 \cdot (1 - |d - 0.5|)$ while TDD utilization is $100 \cdot (1 - g)$. TDD is better when the guard fraction is smaller than the distance of the traffic from 50:50, that is, when $|d - 0.5| > g$. The break-even points are therefore:

- 2 % guard: TDD wins beyond about 52:48.
- 5 % guard: TDD wins beyond about 55:45 (it already wins clearly at 70:30).
- 10 % guard: TDD wins beyond about 60:40.

FDD remains competitive, and is slightly better, only for nearly symmetric traffic, within about $\pm g$ of 50:50 (for example the 50:50 case, where FDD gains 5 points over TDD with a 5 % guard). TDD's advantage is largest for the most asymmetric traffic tested: at 90:10 it is 35 percentage points ahead of FDD with a 5 % guard (38 with 2 %, 30 with 10 %).

D. FDD/TDD trade-offs from Section 1.8

Table 2.4: How the simulation relates to the Section 1.8 trade-offs

Trade-off	Effect observed or expected	In the simulator
Paired-spectrum requirement	FDD needs two separated bands; TDD works in one unpaired band, which is easier to find at high frequencies.	Both schemes are given the same total bandwidth; the requirement itself is not simulated.
Synchronization needs	TDD base stations must share frame timing (GPS-disciplined clocks) to avoid cross-cell DL/UL interference; FDD needs none.	Not simulated; it is an operational cost in addition to the guard overhead.
Guard-period overhead	Time spent switching direction carries no traffic and adds a small latency floor.	Modelled directly as a 2 %, 5 % or 10 % loss of usable capacity.
Slot-ratio flexibility	TDD can match its DL/UL ratio to traffic; FDD's 50/50 split is fixed, so spare capacity is wasted.	TDD follows the demand ratio; FDD loses 20–40 points at 70:30 to 90:10.

Limitations. The model is idealised: the TDD slot ratio is assumed to match the demand exactly (a real frame has a whole number of slots and a limited set of standard UL/DL configurations), the guard overhead is expressed as a fraction of frame time rather than as individual slots, and the multi-frame time-series visualisation (optional in the manual) was not implemented.

E. Program output

Figure 2.3: Terminal output of fdd_tdd_simulator.py (parameters, results table, FDD/TDD interpretation)

```
PS D:\MOBILE_COMMUNICATIONS_MINI_PROJECTS\02_FDD_TDD> python fdd_tdd_simulator.py
```

FDD VS TDD RESOURCE UTILIZATION

System Parameters:

```
Total capacity: 100.0
FDD DL capacity: 50.0
FDD UL capacity: 50.0
TDD guard overhead: 5.0%
```

Main Results:

Scenario	DL_Demand_%	UL_Demand_%	FDD_DL_Throughput	FDD_UL_Throughput	FDD_Total_Throughput	FDD_Utilization_%
50:50	50.0	50.0	50.0	50.0	100.0	100.0
70:30	70.0	30.0	50.0	30.0	80.0	80.0
80:20	80.0	20.0	50.0	20.0	70.0	70.0
90:10	90.0	10.0	50.0	10.0	60.0	60.0

Main results saved to:
results.csv

TDD GUARD-PERIOD ANALYSIS

Scenario	DL_Demand_%	UL_Demand_%	Guard_Overhead_%	TDD_DL_Throughput	TDD_UL_Throughput	TDD_Utilization_%
50:50	50.0	50.0	2.0	49.0	49.0	98.0
70:30	70.0	30.0	2.0	68.6	29.4	98.0
80:20	80.0	20.0	2.0	78.4	19.6	98.0
90:10	90.0	10.0	2.0	88.2	9.8	98.0
50:50	50.0	50.0	5.0	47.5	47.5	95.0
70:30	70.0	30.0	5.0	66.5	28.5	95.0
80:20	80.0	20.0	5.0	76.0	19.0	95.0
90:10	90.0	10.0	5.0	85.5	9.5	95.0
50:50	50.0	50.0	10.0	45.0	45.0	90.0
70:30	70.0	30.0	10.0	63.0	27.0	90.0
80:20	80.0	20.0	10.0	72.0	18.0	90.0
90:10	90.0	10.0	10.0	81.0	9.0	90.0

Conclusion

The simulation shows that fixed FDD allocation leaves capacity unused as soon as traffic is asymmetric: utilization drops from 100 % at 50:50 to 60 % at 90:10, because the saturated downlink cannot borrow idle uplink capacity. TDD keeps about 95 % utilization in every scenario because its time split follows demand, and pays only the guard overhead. TDD overtakes FDD once the traffic is more asymmetric than about 55:45 (5 % guard), and a larger guard overhead moves this break-even towards more asymmetric traffic. FDD remains the better or equal choice for near-symmetric traffic and avoids

synchronization requirements, whereas TDD gives the largest benefit for strongly downlink-dominated traffic such as video streaming.

Mini Project 1.3 — Validating and Extending the 0G–6G Generation Comparison

Project Title

Mini Project 1.3: Validating and Extending the 0G–6G Generation Comparison (fact-check and keyword-based generation recommender).

Objective

To validate the generation summary table of Section 1.2 against additional credible sources, identify the nuance it omits (especially peak versus typical data rates), identify the problems that motivated each generational transition, and build a small rule-based Python recommender that maps a communication description to the generation that first addressed it.

Problem Statement

The Section 1.2 table compresses decades of mobile technology into a few rows of access technology, primary service and peak data rate. Such a table can hide regional differences, multiple standards per generation, and the fact that headline peak rates are rarely achieved in practice. The project verifies the table's claims against independent sources and builds an artifact that demonstrates which generation solved which class of problem.

Tools and Software Used

- Research sources: ITU-R recommendations and press releases, 3GPP/GSA material, vendor technical notes and industry articles (listed in the References).
- Python 3 (script `generation_recommender.py`) with plain string matching for the recommender; pandas (and `openpyxl`) only to export the research table to `generation_research.csv` and `research_table.xlsx`.
- VS Code terminal, Git and GitHub. No infographic drawing tool was needed because the recommender option was chosen.

Background

The broad evolution runs from analogue cellular voice (1G), through digital voice and SMS (2G), mobile data and web access (3G), all-IP mobile broadband (4G), to 5G with its three service classes (eMBB, URLLC, mMTC). The ITU-R defines each generation through a framework: IMT-2000 (3G), IMT-Advanced (4G), IMT-2020 (5G) and IMT-2030 (6G). These frameworks state requirements or targets, which are not the same as speeds measured by users in the field.

Procedure Used

1. List the claims of the Section 1.2 table (access technology, primary service, approximate peak data rate, era) for every generation.
2. Cross-check every generation row against at least two independent sources beyond the unit notes and record the numbers found and why they differ.
3. Identify at least two problems that motivated each transition from one generation to the next.

4. Choose the Python recommender as the deliverable and design a dictionary of generations and keyword lists.
5. Implement `recommend_generation()`: count the keywords of each generation found in the text, return the generation with the highest score, return “Ambiguous” on a tie and “not detected” on zero matches.
6. Test it with seven statements (at least six required), including one that does not map to a generation, and save them in `test_results.txt`.
7. Write the synthesis and discuss where real-world data nuances the unit’s table.

Exact Observations

A. Annotated fact-check of the Section 1.2 table

Table 3.1: Fact-check of the generation table against external sources (reference numbers refer to the References section)

Gen.	Claim in Section 1.2	Finding from external sources	Verdict and nuance
1G	FDMA analogue, voice only, ~1980s.	NTT launched the first commercial cellular network in Tokyo in 1979 and AMPS followed in Chicago in 1983; NMT (1981) was the first with international roaming [25][26].	Confirmed technically. The era is approximate: 1G began in 1979, so “1980s” and “0G before 1979” are rounded.
2G	TDMA/CDMA digital voice and SMS; ~50–200 kbps with 2.5G/2.75G.	GPRS peaks at 171.2 kbps; EDGE roughly triples this to 384 kbps [19]. Real-life ranges are about 35–171 kbps for GPRS and 120–384 kbps for EDGE [20].	Partly outdated: EDGE’s peak exceeds the 200 kbps upper bound; typical GPRS throughput is far below its peak.
3G	WCDMA; ~2–14 Mbps with HSPA.	HSDPA reaches 14.4 Mbps (Rel-5); HSPA+ reaches 21 Mbps (Rel-7), 42 Mbps (Rel-8) and 84 Mbps (Rel-9) [21][22]. Typical HSPA averages are only about 1–3 Mbps [20].	Confirmed for HSPA, but HSPA+ exceeds the range and typical speeds are several times lower than the peaks.
4G	OFDMA/SC-FDMA; ~100 Mbps–1 Gbps.	IMT-Advanced targets are 100 Mbps for high mobility and 1 Gbps for low mobility [23]. First-release LTE did not meet them; LTE-Advanced and WirelessMAN-Advanced were later accorded the IMT-Advanced designation [23][24].	1 Gbps is a requirement for LTE-Advanced, not what a typical LTE user receives; “4G” was also used commercially for first-release LTE.
5G	OFDMA (5G NR); up to ~10–20 Gbps theoretical peak.	IMT-2020 minimum requirements: 20 Gbps downlink and 10 Gbps uplink peak (ideal conditions), but only 100 Mbps downlink and 50 Mbps uplink user-experienced rate at the 5th percentile; connection density of one million devices per km ² [13][14].	Confirmed: the table’s 10–20 Gbps matches the UL/DL peaks, but user-experienced rates are about 200 times lower.
6G	Under study; target 100+ Gbps.	ITU-R M.2160 gives peak-rate examples of 50, 100 and 200 Gbps depending on the scenario, as research targets under ideal conditions; user-experienced targets of 300–500 Mbps are suggested [15][16][17]. Radio-interface proposals are expected from early 2027 and standards by about 2030 [18].	Consistent, but “100+ Gbps” is one of three example values and not a measured result; 6G figures are goals, not field data.

Note on source quality. The 1G facts [25][26] come from general-reference sources and should be complemented by a textbook or ITU history source; the 2G/3G typical-speed figures [19][20] are indicative ranges and not guaranteed values.

B. Problems that motivated each generational transition

Table 3.2: At least two motivating problems per transition

Transition	Problem 1	Problem 2 (and 3)
0G → 1G	Very few channels per city and no frequency reuse, so capacity was tiny.	High-power single transmitters and (in early versions) manual operator connection did not scale; the cellular concept solved both.
1G → 2G	Fragmentation: incompatible national standards (AMPS, NMT, TACS, C-Netz, Radiocom 2000) prevented roaming; GSM was designed as one standard.	No encryption; ESN/MIN sent in clear allowed eavesdropping and cloning fraud, so GSM added SIM authentication and A3/A8/A5 encryption. Analogue also gave low capacity.
2G → 2.5G / 3G	Circuit-switched data wasted capacity for bursty traffic; GPRS introduced packet switching and volume-based charging.	Rates of only tens to hundreds of kbps [19] could not support mobile web and multimedia, motivating IMT-2000 and HSPA.
3G → 4G	Typical HSPA speeds of a few Mbps [20] and higher latency limited video and apps.	Separate circuit-switched voice core and RNC-based architecture; 4G adopted an all-IP flat core (EPC), VoLTE, OFDMA and MIMO for capacity.
4G → 5G	Latency and reliability were insufficient for real-time control (autonomous vehicles, remote surgery, industrial automation), motivating URLLC (~1 ms).	IoT scale of up to one million devices per km ² [14] (mMTC) and demand for multi-Gbps rates (eMBB, mmWave and mid-band spectrum).
5G → 6G	Need for AI-native network operation and integrated sensing and communication.	Coverage gaps and the need for terrestrial-satellite integration, plus higher peak-rate targets of 50–200 Gbps [15].

C. Recommender tool: design

The deliverable is a rule-based Python program (`generation_recommender.py`). It stores, for each generation 0G–6G, its period, technology, features, examples, services and the main problem that led to the next generation, together with a keyword list per generation (for example “sms” and “gsm” for 2G, “lte” and “volte” for 4G, “urllc” and “massive iot” for 5G, “ai-native” and “integrated sensing” for 6G). For an input text it lowercases the text, counts how many keywords of each generation occur, and returns the generation with the highest count; equal best scores return “Ambiguous”, and no match returns “not detected”. Test statements are run automatically and saved to `test_results.txt`, and an interactive mode lets the user type any device or technology. The research table is also exported to `generation_research.csv` and `research_table.xlsx`.

D. Recommender tool: test cases and results

Table 3.3: Test statements and outputs (from `test_results.txt`)

#	Statement	Recommendation	Comment
1	This system used analog voice communication.	1G	Correct (keyword “analog voice”).
2	SMS and digital voice became important services.	2G	Correct (“sms”, “digital voice”).
3	Mobile Internet and video calling became available.	3G	Correct (“mobile internet”, “video calling”).
4	LTE provides high-speed mobile broadband.	4G	Correct (“lte”, “mobile broadband”).
5	5G supports massive IoT and ultra-reliable low-latency communication.	5G	Correct (“5g”, “massive iot”).
6	Future networks may use AI-native communication and integrated sensing.	6G	Correct (“ai-native”, “integrated sensing”, “future network”).
7	Mobile communication provides faster Internet.	Ambiguous / Unknown	No generation-specific keyword; this is the case that does not map to a single generation.

Figure 3.1: Terminal output of `generation_recommender.py` showing the seven automatic tests

```

PS D:\MOBILE_COMMUNICATIONS_MINI_PROJECTS\03_0G_6G> python generation_recommender.py
=====
0G-6G GENERATION RECOMMENDER
=====

Test 1
Statement: This system used analog voice communication.
Recommendation: 1G
Confidence: Keyword-based
Reason: Matched generation-specific keywords for 1G.

Test 2
Statement: SMS and digital voice became important services.
Recommendation: 2G
Confidence: Keyword-based
Reason: Matched generation-specific keywords for 2G.

Test 3
Statement: Mobile Internet and video calling became available.
Recommendation: 3G
Confidence: Keyword-based
Reason: Matched generation-specific keywords for 3G.

Test 4
Statement: LTE provides high-speed mobile broadband.
Recommendation: 4G
Confidence: Keyword-based
Reason: Matched generation-specific keywords for 4G.

Test 5
Statement: 5G supports massive IoT and ultra-reliable low-latency communication.
Recommendation: 5G
Confidence: Keyword-based
Reason: Matched generation-specific keywords for 5G.

PS D:\MOBILE_COMMUNICATIONS_MINI_PROJECTS\03_0G_6G> python generation_recommender.py
Now you can enter a device, technology, service, or description.

Examples:
Nokia 3310
GSM phone with SMS
LTE smartphone
5G IoT sensor
analog mobile phone
video calling
AI-native network
integrated sensing
push to talk radio

Type 'list' to see all generations.
Type 'exit' to finish.

Enter device or communication technology: sms

Detected generation: 2G

=====
INFORMATION FOR 2G
=====
Period      : 1990s
Technology  : Digital cellular / GSM
Features    : Digital voice, SMS and improved capacity
Examples    : GSM phones, Nokia 3310
Services    : Voice, SMS and basic data
Main problem : Limited data rates for Internet applications
=====

Enter device or communication technology: 

```

Figure 3.2: Interactive search with one example input (for example “Nokia 3310”)

E. Where real-world data nuances the unit’s table

The clearest case is the 5G row. The table’s “up to ~10–20 Gbps” agrees with the ITU peak requirements of 20 Gbps downlink and 10 Gbps uplink [13][14], but those are ideal-condition maxima per device. The same ITU document requires only 100 Mbps downlink and 50 Mbps uplink as the experienced rate of the 5th-percentile user in dense urban areas, about 200 times lower. The same pattern appears in the 2G row (EDGE peak of 384 kbps but a real-life range of 120–384 kbps [19][20]), in the 3G row (HSPA+ peaks of 21–84 Mbps against averages of 1–3 Mbps [20][21]), in the 4G row (1 Gbps was a requirement that first-release LTE did not meet [23][24]) and in the 6G row (50–200 Gbps are research targets [15]). The 1G row is nuanced by dates: the first commercial cellular network started in Japan in 1979 [25], earlier than the “1980s” of the table.

F. Limitations of the recommender

The tool matches technology, service and device names and therefore identifies the generation of a described system. Keyword matching with simple substring tests cannot understand meaning, can match a keyword inside a longer word (for example “edge” or “lte”), and misses synonyms that are not in the dictionary. The tests mostly use technology names; requirement-style sentences such as “low-latency control for a robotic arm” would need extra problem-class keywords (for example “low latency”, “robot”, “remote surgery” for 5G URLLC) to be recognised.

Conclusion

The research confirms the broad progression of the Section 1.2 table but shows that a generation is not a single technology and that headline rates are targets or ideal-condition peaks rather than typical user speeds: EDGE exceeds the table’s 2G range, HSPA+ exceeds its 3G range, 1 Gbps was an IMT-Advanced requirement rather than an LTE norm, 5G’s 20 Gbps peak contrasts with a 100 Mbps experienced rate, and 6G figures are still research goals. The identified problems (fragmentation and security weaknesses, circuit-switched data inefficiency, low latency and IoT scale, AI-native sensing) explain why each generation was created. The Python recommender maps descriptions to generations through keyword matching and passed all seven tests, including the case that does not map to a single generation, but needs requirement-style keywords to become a true problem-to-generation tool.

Overall Conclusion

The three projects examine propagation, resource allocation and technological evolution. Project 1.1 demonstrated that frequency and distance determine received power and coverage: with identical transmit parameters, 28 GHz suffers 32 dB more path loss than 700 MHz, so its range is about 40 times shorter, and bandwidth is a far more effective way to raise capacity than power. Project 1.2 demonstrated that duplexing architecture matters under asymmetric traffic: FDD wastes idle capacity (60 % utilization at 90:10) while TDD keeps about 95 % and wins once traffic is more asymmetric than the guard-overhead break-even, at the cost of guard time and synchronization. Project 1.3 demonstrated that mobile generations evolved to solve concrete problems and that generation tables must be checked against independent sources because peak rates, typical rates and requirements differ.

Requirements Compliance Summary

Project	Requirement of the lab manual	Where addressed
1.1	FSPL and link-budget functions; two bands with identical parameters; plot with sensitivity line and legend; maximum range and link margin per band; saved figure.	Procedure; Table 1.2; Figure 1.1
1.1	Lab Tasks 1–6 (24 dBi gain, 3.5 GHz, 8 dB fading, SNR/capacity, power versus bandwidth, TDD duty-cycle note).	Sections C–F of Exact Observations; Tables 1.3–1.5; Figures 1.2–1.3
1.1	Written analysis of the range difference in terms of the free-space path-loss formula.	Exact Observations, section B
1.2	FDD model (two fixed channels) and TDD model (shared channel, guard overhead, DL/UL split); demand given as DL:UL ratio; at least three scenarios.	Background model; Table 2.2 (four scenarios)
1.2	Throughput per direction and utilization for both schemes; chart or table; at least two guard fractions.	Tables 2.2–2.3; Figures 2.1–2.2 (2 %, 5 %, 10 %)
1.2	Asymmetry beyond which TDD wins; range where FDD stays competitive; Section 1.8 trade-offs; when TDD's advantage is largest.	Sections C and D of Exact Observations
1.2	Optional multi-frame time-series visualisation.	Not implemented (optional)
1.3	Cross-check at least four generations against at least two additional sources each; discrepancies noted.	Table 3.1 (all six generations)
1.3	At least two problems per generational transition.	Table 3.2
1.3	Recommender program with at least six varied tests, including one that does not map to a single generation; results recorded.	Sections C–D; Table 3.3 (seven tests)
1.3	Written report tying research and deliverable together, with at least one case of real-world data nuancing the table.	Section E and Conclusion

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GitHub Link

Repository: <https://github.com/IRAFASHA-Justine/MOBILE-COMMUNICATION-WORKS->

